SUPPLEMENTAL METHODS

Study Population

The study is based on a cohort of 189 initially caries-free, Medicaid and Child Health Plus eligible preschool children and their primary caregivers, drawn from patients 1-3 years of age presenting to the Division of Pediatric Dentistry in Eastman Institute for Oral Health (EIOH) for regular dental visits. The study was approved by the University of Rochester Medical Center Research Subject Review Board (RSRB#0000759) and written informed consent was obtained from parents/primary caregivers of participating children prior to the study. Children were followed for up to 24 months. At baseline and at each 6-month follow-up visit, a calibrated dental caries examiner (kappa statistics 80-95%) assessed the oral health status of the child and collected a saliva sample from the child. ECC was defined in accordance with American Academy of Pediatric Dentistry Guidelines [decayed, missing (due to caries) and filled (due to caries) surfaces > 0] (AAPD 2011). An International Caries Detection Assessment System, (ICDAS) was used as a caries index (Ismail et al. 2007). For this study, we selected 36 children with ECC onset that were among the first to become caries-active during the 24-month follow-up period. Pre-caries samples were analyzed for all of these children and caries-active samples were analyzed for all but three of them. We matched these 36 subjects with a random sample of 20 children who remained caries-free for the 24-month study period. The selection criteria for these 20 children included comparable carriage of *Streptococcus mutans*, *Lactobacillus*, and *Candida* at baseline and matching by age, race, and ethnicity. Samples from persistently caries-free healthy controls were analyzed through the first three visits. Note that culture results from the 56 subjects showed significant difference in in carriage of *S. mutans* ($p = 0.008$) but not in carriage of *Lactobacillus*, ($p = 0.37$), *Candida* ($p = 0.24$) or total CFU ($p = 0.37$).

Sample collection

At each study visit, whole, stimulated, saliva samples were obtained through a disposable saliva ejector attached to a 50 ml sterile centrifuge tube, which in turn was attached to a vacuum pump (Mundorff et al. 1990). Approximately 2 ml of saliva was collected for each sample. The saliva sample was kept on ice and then transferred to a cyrotube and frozen at -80C. For the microbiome analyses, the samples that were selected were thawed and 50ul was transferred to a new sterile microfuge tube and frozen at -80C.

Microbiology laboratory procedures

All laboratory analytical procedures were completed without knowledge of caries status. At the time of collection, samples were identified only by date, examination number, and subject ID. Samples were processed between 1 and 4 hours after collection, as there is no significant loss of numbers of total viable flora during the first 24 hours (Mundorff et al. 1990). For dispersion of cell clumps and dechaining of streptococci, a 1 ml aliquot of saliva was sonicated (three 10-second sonic bursts, 100 Watts of peak power). The suspension was serially diluted (10-fold dilutions) in phosphate buffer and 100 ml aliquots uniformly plated on: (i) Mitis Salivarius (MS) agar (Beckson- Dickinson) supplemented with 20% sucrose and bacitracin (0.2 U/ml), to determine the presence of MS, (ii) Rogosa agar (Difco) to evaluate the presence of Lactobacilli, (iii) Saboraud dextrose (Oxoid) supplemented with 0.1 mg/ ml chloramphenicol to determine the presence of *Candida* spp., and (iv) tryptic soy agar supplemented with 5% sheep blood to enumerate the total microflora. MS plates were incubated at 37°C in a 5% CO_2 atmosphere, Rogosa plates were incubated anaerobically at 37°C, Saboraud plates were incubated aerobically at 37°C and duplicated blood agar plates incubated at 37°C both under aerobic and anaerobic conditions. All dilutions were plated in triplicates and the plates incubated under the described conditions for 72 h before colonies were counted. The number of MS, Lactobacilli, *Candida* spp. and total microflora was expressed as colony forming unit (CFU) per ml of saliva.

Genomic DNA extraction

Total genomic DNA was extracted from the nasal samples using a modification of the ZymoBIOMICS^{TM} DNA Miniprep Kit (Zymo Research, Irvine, CA) and FastPrep mechanical lysis (MPBio, Solon, OH). 16S ribosomal DNA (rRNA) was amplified with Phusion High-Fidelity polymerase (Thermo Scientific, Waltham, MA) and dual indexed primers specific to the V1-V3 (8F: 5' AGAGTTTGATCCTGGCTCAG 3'; 534R: 3' ATTACCGCGGCTGCTGG 5') hypervariable regions as previously described (Grier et al. 2018). Amplicons were pooled and paired-end sequenced on an Illumina MiSeq (Illumina, San Diego, CA) in the University of Rochester Genomics Research Center. Positive controls comprised of mock communities and negative controls consisting of sterile saline were processed in parallel to the biological samples were used to validate the sequencing data and verify the absence of contamination, which was actively mitigated by UV irradiation of all reagents and sample processing equipment.

Microbiota background control

The background microbiota was monitored at multiple stages of sample collection and processing. All sterile saline, buffers, reagents, plasticware and flocked nylon swabs used for sample collection, extraction and amplification of nucleic acid were UV irradiated to eliminate possible DNA background contamination. Elimination of potential background from the irradiated buffers, reagents, plasticware and swabs was confirmed by 16S rRNA amplification. After sample collection, multiple aliquots of sterile saline with swabs used for sample collection were carried through our entire sequencing protocol as individual samples, including DNA extraction, 16S rRNA amplification, library construction and sequencing to monitor potential background microbiome. Data from these background control samples is deposited in SRA along with positive controls.

Bioinformatics and statistical analysis

Raw data from the Illumina MiSeq was first converted into FASTQ format 2x312 paired end sequence files using the bcl2fastq program, version 1.8.4, provided by Illumina. Format conversion was performed without de-multiplexing and the EAMMS algorithm was disabled. All other settings were

default. Reads were multiplexed using a configuration described previously (Fadrosh et al. 2014). Briefly, for both reads in a pair, the first 12 bases were a barcode, which was followed by a primer, then a heterogeneity spacer, and then the target 16S rRNA gene sequence. QIIME 1.9.1 (Caporaso et al. 2010) was used to extract the barcodes into a separate file for importing into QIIME 2 V2018.11) (Bolyen et al. 2019), which was used to perform all subsequent processing. Reads were demultiplexed requiring exact barcode matches, and 16S rRNA primers were removed allowing 20% mismatches and requiring at least 18 bases. Cleaning, joining, and denoising were performed using DADA2(Callahan et al. 2016): forward reads were truncated to 275 bps and reverse reads to 260 bps, error profiles were learned with a sample of one million reads, and a maximum expected error of two was allowed. Taxonomic classification was performed with a custom naïve Bayesian classifier trained on the August, 2013 release of GreenGenes (DeSantis et al. 2006; McDonald et al. 2012). Sequence variants that could not be classified at least at the phylum level were discarded. 16S rRNA gene amplicon sequence variants observed fewer than ten times total, or in only one sample, were discarded. Samples with fewer than 11,500 reads were discarded. Phylogenetic trees were constructed for each cohort using MAFFT (Katoh and Standley 2013) for sequence alignment and FastTree (Price et al. 2010) for tree construction. Alpha diversity was computed using the Shannon and Faith's metrics and beta diversity was computed using the weighted Unifrac metric (Lozupone et al. 2011). All diversity metrics were calculated on samples rarefied to 11,500 reads using QIIME 2.

Comparisons of alpha diversity between groups was performed with the Wilcoxon rank sum test and association with age was assessed using Spearman correlation. Beta diversity differences between groups were tested with PERMANOVA (Anderson 2001). Univariate tests for differentially abundant microbes was performed with the Wilcoxon rank sum test via SIAMCAT (Wirbel et al. 2020): samples were filtered with a minimum prevalence threshold of 0.1, centered log ratio normalization was performed with a pseudocount of $1x10^{-6}$, and associations were tested on normalized data at an alpha of 0.05. Multivariate differential abundance between caries status groups was assessed using Maaslin2 (Mallick et al. 2019): minimum feature abundance was $1x10^{-6}$, minimum prevalence was 0.1, normalization was total sum scaling, transformation was arcsin square root, fixed effects were caries status, age, and sequencing

reads, and subject was used as a random effect. Random forest and gradient boosting caries risk classification models were trained on the abundances of all species and tested using eight-fold nested cross-validation via the QIIME 2 classify-samples-ncv from the sample-classifier plugin with 5,000 estimators each and automatic tuning of hyperparameters using random grid search(Bokulich et al. 2018; Pedregosa et al. 2011). This procedure was run for each type of model twice: once using all enrollment samples and once using all pre-caries samples which immediately preceded a caries active visit plus a single sample from each healthy control subject taken from the third visit. Receiver operating curves, confidence intervals, and the areas under the curve (AUC) were calculated with the pROC R package(Robin et al. 2011). AUC p-values were calculated with the verification R package and tested the null hypothesis that the AUC was <= 0.5. Important features were visualized with SIAMCAT (Wirbel et al. 2020).

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Supplemental Table 1. Demographic table for the study cohort of 56 children ages one to four years and caries-free at the time of enrollment*.

Variable	Converted to dental caries (n=36)	Remained Caries-free (n=20)	p-value
Gender			0.04 [2]
Female	13	13	
Male	23	7	
Ethnicity			0.86 [2]
Hispanic	10	6	
Not Hispanic	26	14	
Race[1]			0.68 [3]
Native American	1	0	
African American	15	7	
Mixed	13	10	
Caucasian	7	2	
Baseline Candida[1]			0.36[2]
Positive	11	4	
Negative	24	16	

[1]Information not available for one subject
[2]Pearson's chi-square test
[3]Fisher's exact text
[*]Subjects were followed for 24 months after enrollment and were subsequently divided into two groups based on whether or not they converted to a state of active caries during the study period.